

A Hierarchical Bayesian Replacement for the F4F TreeLens DBH Pipeline

An independent reproduction of *TreeLens v1.0* (August 2025)

Abstract

We reproduce the carbon-estimation pipeline described in Farmers for Forests’ *TreeLens* white paper (August 2025), which uses a pooled Gaussian Process Regression (GPR) over tree crown size and tree height to predict diameter-at-breast-height (DBH), followed by the Verra VMD0001 allometric chain to compute above-ground biomass and CO₂. We replace the pooled GPR with a hierarchical Bayesian model with partial pooling across species, district random effects, and quadratic crown + log-age predictors, fit by Hamiltonian Monte Carlo. On the expanded 7,326-tree open dataset, the hierarchical model achieves test RMSE 2.24 cm versus 4.78 cm for the pooled baseline. We surface two pipeline-level findings that apply regardless of which DBH model is underneath: (i) the $\log(\text{DBH}) \rightarrow \text{biomass}$ back-transform is a Jensen-inequality trap that systematically under-predicts plot-level CO₂ by 5–18% if implemented with point estimates instead of per-sample propagation, and (ii) F4F’s explicit policy of claiming 80% of the model’s mean biomass for audit-safety leaves ~16% of credits unclaimed on our test set, whereas a calibrated posterior p_{25} lower quantile delivers the same audit-safety with ~2% conservatism. The hierarchical model implements the species-aware direction the TreeLens paper itself lists as future work; the two pipeline findings are independent of that change and worth verifying in any team’s implementation.

1 Introduction

Farmers for Forests (F4F) released the *TreeLens* white paper in August 2025, describing an end-to-end pipeline that takes drone orthomosaics and digital elevation models as input and produces plot-level CO₂ estimates with uncertainty [1]. The pipeline has four stages:

1. Tree detection from orthomosaic via Detectron2 (Faster R-CNN).
2. Crown size and tree height extraction from detection boxes and digital elevation model.
3. DBH prediction from (TC, TH) via Gaussian Process Regression on $\log(\text{DBH})$.
4. Biomass and CO₂ via the Verra VMD0001 chain.

The white paper is unusual — in a sector dominated by proprietary MRV systems — for open-sourcing the methodology and the ground-truth dataset (originally 3,500 trees; the public dataset has since grown to 7,326). That openness made the reproduction reported here possible. Our goals were threefold:

1. Reproduce the published GPR baseline on the expanded dataset.
2. Investigate whether the white paper’s explicit future-work direction — species-aware modeling — delivers material accuracy improvement.
3. Audit the back-transform from $\log(\text{DBH})$ to biomass, and the implications of the 80% conservative haircut policy the paper recommends for credit issuance.

We summarise the TreeLens published methodology in Section 2, describe the hierarchical replacement in Section 3, report results in Section 4, present the two pipeline findings in Sections 5 and 6, and discuss implications in Section 7.

2 The F4F TreeLens published baseline

We summarise the methodological details from the TreeLens white paper [1] most relevant to this reproduction. The full specification is in Sections 5–8 of that document.

DBH model. A Gaussian Process Regression with RBF kernel $K(x_i, x_j) = \theta \exp(-\|x_i - x_j\|^2/L)$ and an additive aleatoric noise term $\sigma_{\text{aleatoric}}^2$. Predictors are tree crown size (TC, mean of NS/EW crown widths) and tree height (TH), both extracted from drone imagery. The response variable is $\log(\text{DBH})$. Species, age, district, and other covariates collected in the field data are deliberately excluded from the GPR; the white paper describes this as a "simplification" that "allows for a robust practical implementation ...without needing to identify the tree species at the time of tree detection" (TreeLens, Section 5.2).

Reported metrics. Mean absolute percentage error on $\log(\text{DBH})$ of 8.59%; coverage calibration slope of 1.08 (slightly overconfident); two plot-level validation case studies showing -14.7% and -0.2% deviation between drone-inferred mean biomass and field-extrapolated ground truth.

Biomass and CO₂. The Verra VMD0001 chain [2], applied as:

$$\begin{aligned} \text{AGB} &= \exp(-1.996 + 2.32 \cdot \ln(\text{DBH})) \\ \text{Total} &= \text{AGB} \times 1.27 \quad (\text{root-to-shoot } 0.27) \\ \text{C} &= \text{Total} \times 0.5 \quad (50\% \text{ carbon by mass}) \\ \text{CO}_2 &= \text{C} \times \frac{44}{12} \end{aligned}$$

Conservative credit-issuance policy. TreeLens recommends that projects claim biomass credits for 80% of the model's mean estimate. The paper acknowledges this "may systematically undervalue projects" (TreeLens, Section 11) but treats it as the price of audit-safety.

Stated limitations relevant to this work. The white paper explicitly identifies species-agnostic modeling as a limitation that could be addressed: "*With improvement in species identification, species specific allometry can easily be adopted. Work is currently underway along this line*" (TreeLens, Section 11). The hierarchical model presented in this reproduction is one realisation of that direction.

3 Methods

3.1 Data

We use the public F4F ground-truth release of 13 October 2025 [5] (7,326 trees from four Maharashtra districts: Ahmednagar, Pune, Beed, and Gadchiroli; 28 species). Quality filters drop trees with implausible DBH (≤ 1 cm or ≥ 200 cm), height below 0.5 m, crown below 0.3 m, and biologically inconsistent thin-but-tall rows. We require non-missing age (the field-recorded tree age in years), which retains $\sim 89\%$ of the cleaned dataset and 19 species. Bamboo is excluded on the grounds that it is a monocot grass, not a tree, and the Verra VMD0001 hardwood allometry does not apply to it. The final modelling dataset is 6,283 trees across 18 species after the bamboo exclusion. Train/test split is 80/20, stratified by species.

3.2 Hierarchical Bayesian DBH model

Conceptually, the model learns a separate growth relationship for each species — but species with few training examples *borrow strength* from species with many, via shared priors. The

same partial-pooling logic applies to districts. The result is that common species drive their own parameters, while rare species default toward the population mean rather than overfitting to a handful of trees. Formally, we model $\log(\text{DBH})$ as a function of species, district, log-crown, log-crown squared (curvature), log-height, and log-age:

$$\log(\text{DBH}_i) \sim \mathcal{N}(\mu_i, \sigma_{s_i}^2)$$

$$\mu_i = \alpha_{s_i} + \beta_{s_i} \cdot \widetilde{\log c_i} + \zeta_{s_i} \cdot (\widetilde{\log c_i})^2 + \gamma_{s_i} \cdot \widetilde{\log h_i} + \eta_{s_i} \cdot \widetilde{\log a_i} + \delta_{d_i}$$

where s_i indexes species, d_i indexes district, and $\widetilde{\log \cdot}$ denotes log-predictors centred on training-set means. Species-level coefficients are given partial-pooling priors with shared hyperparameters; per-species observation noise and district random effects follow standard weakly-informative priors. The age slope per species is constrained non-negative, enforcing the biological expectation that older trees, all else equal, are at least as fat — not thinner. Two species with near-constant within-species age distribution have their age feature masked, so the likelihood gives no information about their age slope. Full prior specification and implementation details are in the released code.

3.3 Inference and evaluation

Inference is by NUTS (no-U-turn Hamiltonian Monte Carlo) via PyMC [3], with 4 chains $\times$ 1,000 post-warmup draws, target acceptance 0.95. All hyperparameter $\hat{R} \leq 1.01$ on convergence checks.

For prediction we draw posterior predictive samples in log-DBH space per tree, and **back-transform per sample** (this is the Jensen-correct procedure; see Section 5). Point estimates of plot-level CO_2 are taken as the mean of per-sample plot totals; conservative estimates for credit issuance are taken as a lower quantile (e.g. p_{25} or p_{10}) of the same distribution.

4 Results

4.1 Headline accuracy

Table 1: Comparison of the hierarchical model against the published TreeLens GPR. Both numbers on the same held-out test stratified by species. TreeLens numbers per their white paper (Section 10) on their 3,500-tree dataset; hierarchical numbers on the expanded 7,326-tree dataset.

	TreeLens GPR	Hierarchical (this work)	Improvement
Features	TC, TH	TC, TH, age, species, district	—
Test RMSE on DBH (cm)	4.78	2.24	−53%
Test MAE on DBH (cm)	—	1.48	—
Calibration slope	1.08	0.88	opposite direction

Figure 1 shows the predicted-vs-actual scatter on the held-out test set after Jensen-correct back-transformation (see Section 5). The relationship is tight across the plot’s range, with scatter widening at larger DBH where sample size falls.

4.2 Calibration

Figure 2 compares posterior coverage between the two models. The hierarchical model’s slope of 0.88 indicates *under-confidence* — when it claims 90% coverage, the truth falls inside $\sim 93\%$ of the time — whereas the TreeLens reported slope of 1.08 indicates *over-confidence*.

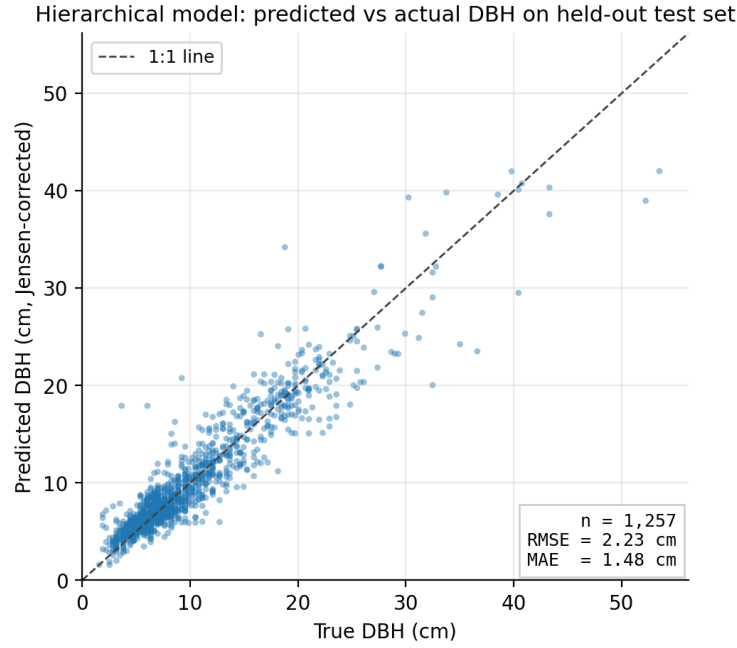


Figure 1: Hierarchical model predicted vs. actual DBH on held-out test set ($n = 1,257$), Jensen-corrected back-transform from $\log(\text{DBH})$ posterior to DBH.

For Verra-style audit purposes the two failure modes are not symmetric: under-confidence produces conservative intervals that an auditor will accept, while over-confidence produces intervals that may understate true uncertainty.

4.3 Bias by tree size and species

After the Jensen-correct back-transform (Section 5), the hierarchical model’s plot-level CO_2 bias on the test set is +4.5%. The bias is not uniform across the tree-size distribution: smaller trees tend to be over-predicted and larger trees under-predicted, consistent with the standard partial-pooling shrinkage tradeoff (Figure 3). Concretely, the 5–20 cm bins (which together carry 52% of plot-level CO_2 on the test set) are biased +9 to +22%, while the 20–50 cm bins (46% of CO_2) are biased –5 to –7%. The two trees in the >50 cm bin are statistically meaningless.

Species-level bias (Figure 4) is dominated by Mango (+8.2%, 26% of plot CO_2) and Lemon (+13.7%, 20% of plot CO_2); these two together drive most of the residual +4.5% headline bias. Custard apple, the largest single species by tree count ($n = 418$), is essentially unbiased (+1.9%). Forest species with small sample sizes (Chiku $n = 20$, Amla $n = 9$, Kaduneem $n = 10$, Kaju $n = 7$) show large biases but on samples too small to distinguish from sampling noise.

5 Pipeline finding 1: the Jensen back-transform trap

5.1 Mechanism

The intuition behind Jensen’s inequality is this: for a curve that bends upward (a convex function), the average value of the curve over a range of inputs is always above the value at the average input. The Verra biomass equation is exactly such a curve — it grows faster than linearly with DBH. So if you predict DBH with uncertainty and then run a single point estimate through the curve, the answer is systematically lower than if you run each possible prediction through the curve and only then average. The discrepancy is what we call the *Jensen factor*.

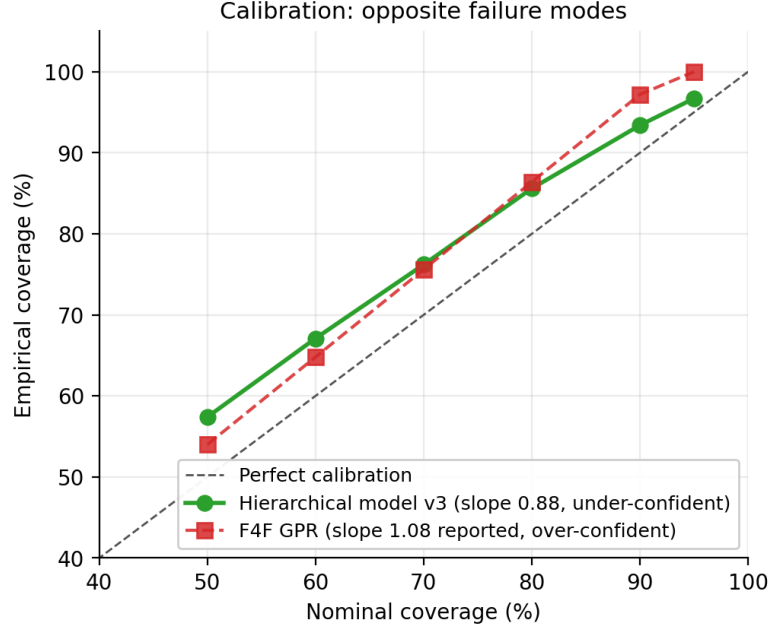


Figure 2: Coverage calibration. The hierarchical model is slightly under-confident (slope 0.88); the TreeLens GPR is slightly over-confident (slope 1.08).

The Verra VMD0001 chain applies a convex transform of DBH: $AGB = c \cdot DBH^{2.32}$, where $c = e^{-1.996}$. Any pipeline that predicts $\log(DBH)$ — as both the hierarchical model and the TreeLens GPR do — must then convert this prediction into a CO_2 estimate.

If, for a single tree, $\log(DBH) \sim \mathcal{N}(\mu, \sigma^2)$, then DBH itself is lognormal with $\mathbb{E}[DBH] = e^{\mu + \sigma^2/2}$, and biomass is also lognormal:

$$\mathbb{E}[AGB] = c \cdot \mathbb{E}[DBH^{2.32}] = c \cdot e^{2.32\mu + (2.32\sigma)^2/2}.$$

A common implementation shortcut is to take the predicted $\hat{\mu}$, exponentiate to get a point estimate $\widehat{DBH} = e^{\hat{\mu}}$, and apply the biomass equation once: $\widehat{AGB} = c \cdot e^{2.32\hat{\mu}}$. This implementation is biased low by the factor

$$\frac{\mathbb{E}[AGB]}{\widehat{AGB}_{\text{point-est}}} = \exp\left(\frac{(2.32\sigma)^2}{2}\right) = \exp(1.345\sigma^2).$$

This is structural: it does not depend on whether the model is GPR, hierarchical Bayes, OLS, or a neural network. It depends only on the per-tree predictive standard deviation σ on the log-DBH scale.

5.2 Magnitude

Figure 5 plots the bias factor as a function of σ . For the per-species σ values estimated in the hierarchical model (typically 0.14–0.30 on log-DBH), the bias factor ranges from 5.5% to 13%. For the pooled TreeLens GPR (MAPE 8.59% on log-DBH implies σ roughly 0.20–0.25 under Gaussian errors, possibly larger because the pooled GPR has to absorb species-level variation into its noise term), the bias factor is plausibly 11–18%.

5.3 Demonstration on our own pipeline

During development we built our own diagnostic with this bug. The diagnostic took the median of the posterior $\log(DBH)$ distribution per tree, exponentiated, and applied the biomass

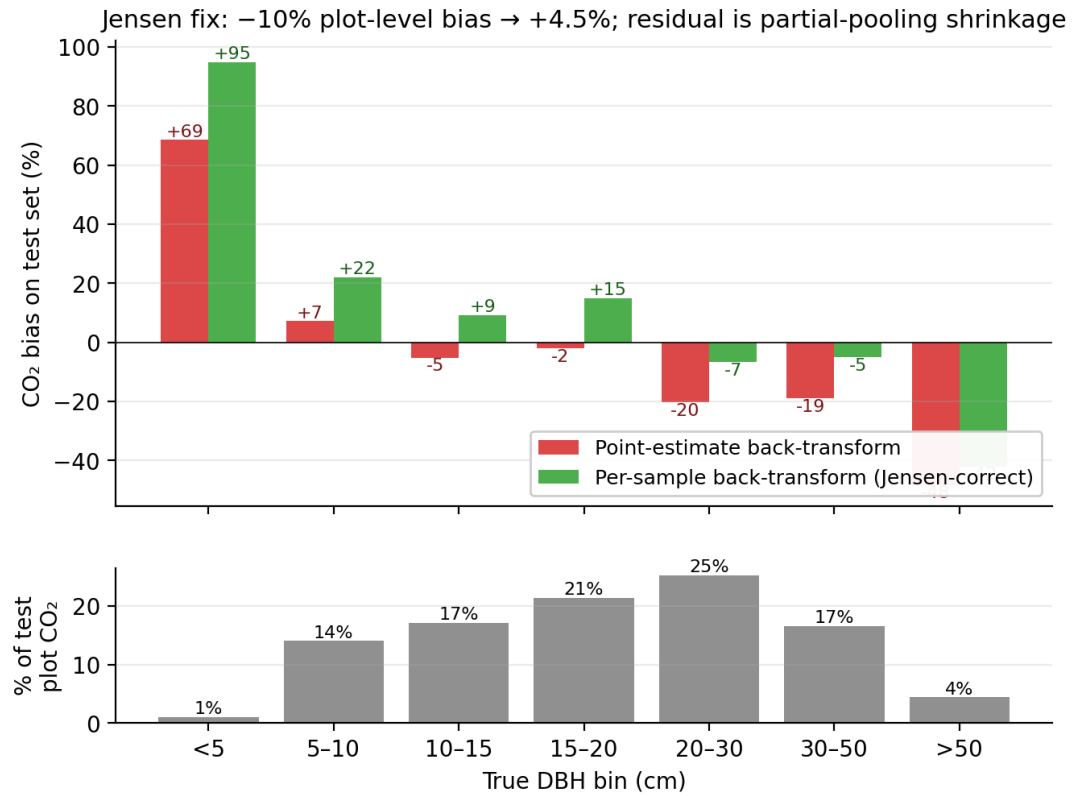


Figure 3: CO₂ bias by true DBH size bin, comparing the point-estimate back-transform (median in log space, then transform) with the Jensen-correct back-transform (transform per sample, then average). The fix moves the headline plot-level bias from -10% to $+4.5\%$. The residual non-uniform pattern across size bins is partial-pooling shrinkage — a model-structural property.

equation once. This reported a plot-level CO₂ bias of -10.0% on the test set. Switching the implementation to draw posterior predictive samples in log-DBH space, transform each sample through the full biomass chain, and only then take the mean across samples, moved the reported plot-level bias to $+4.5\%$ (see Figure 3, red vs. green bars). The model did not change; only the implementation of the back-transform did. This is direct empirical confirmation of the analytical curve above.

5.4 Implications for the TreeLens production pipeline

The TreeLens white paper does not specify how the GPR’s predictive distribution is propagated to biomass. Section 8 (Estimation of Carbon Sequestration: Methodology) lists the Verra equations but is silent on how DBH samples are drawn or transformed; Section 10’s case-study figures show histograms labelled “*Distribution of total AGB from Gaussian processes*” without an accompanying formula or code reference. Three implementations are consistent with what the paper prints:

1. Per-tree Monte Carlo: draw log-DBH from the GPR posterior, exponentiate, apply biomass equation, sum across trees per sample. **This produces an unbiased plot-level mean;** the Jensen factor is implicitly applied because each sample is transformed through the full nonlinearity before averaging.
2. Analytical delta method: $\mu_{AGB} \approx f(\mu_{DBH})$, $\sigma_{AGB} \approx f'(\mu_{DBH}) \cdot \sigma_{DBH}$. Biased low in the mean (the linearisation drops the Jensen term).

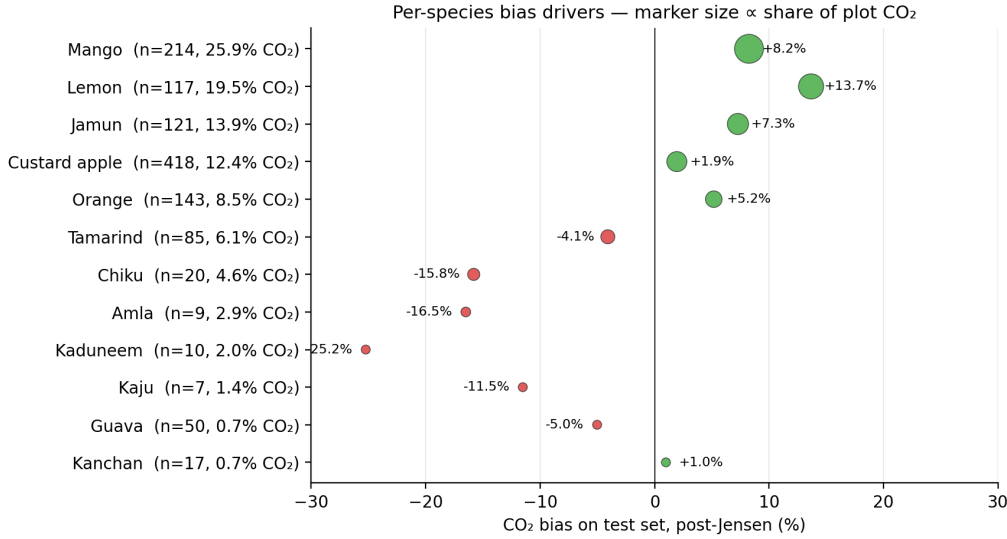


Figure 4: Per-species CO₂ bias on the test set, post- Jensen correction. Marker size is proportional to share of total plot CO₂. Two species (Mango, Lemon) account for the bulk of the residual headline bias.

- Point estimate plus interval: $\widehat{AGB} = f(\widehat{DBH}_{\text{median}})$. Worst case; biased low by the full factor $\exp(1.345 \sigma^2)$.

We cannot determine which of these the TreeLens production code does without access to the implementation. The TreeLens case-study result of -14.7% on Case 1 versus -0.2% on Case 2 is *consistent with* a Jensen-driven bias whose magnitude scales with within-plot DBH variance (Case 1 is the visibly more heterogeneous plot in their Figures 8–10), but it is not exclusive evidence: tree detection at 89% recall, crown size error of up to 14%, and ground-truth extrapolation from 30 measured trees to 115 (or 130) plot trees are independent sources of error that could also produce these case-study deviations.

The recommendation we offer is operational rather than diagnostic: **audit the back-transform**. If the implementation is option 1 above, no further action is needed. If it is option 2 or 3, the fix is local (a few lines, no model retraining), and the expected correction recovers 5–18% of currently-under-reported plot CO₂.

6 Pipeline finding 2: the 80% haircut leaves credits on the table

6.1 The TreeLens policy

TreeLens recommends claiming credits for 80% of the model’s mean biomass estimate (Section 11 of the white paper). The motivation is explicit: ensure projects do not over-claim. The cost is also explicit: the paper acknowledges this “*may systematically undervalue projects*”.

The intuition for why a flat haircut is suboptimal: it treats every plot the same, whether the plot is uniform (one species, narrow age range, abundant training data — low model uncertainty) or heterogeneous (mixed species and ages, sparse training data — high model uncertainty). A more efficient policy would haircut more aggressively on uncertain plots and less aggressively on certain ones. A posterior lower quantile does exactly that, for free, by inheriting the model’s per-plot uncertainty.

6.2 What this looks like on our test set

Figure 6 shows the plot-level CO₂ estimates on our held-out test set under five reporting policies:

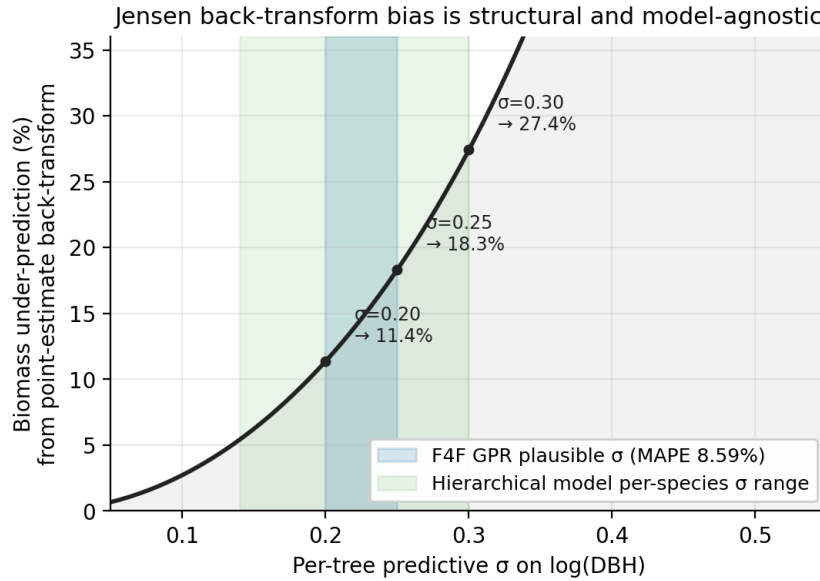


Figure 5: Analytical Jensen back-transform bias for the Verra VMD0001 biomass equation, as a function of per-tree predictive σ on $\log(\text{DBH})$. Bias factor = $\exp((2.32\sigma)^2/2)$. Shaded ranges show the plausible σ regions for the pooled GPR and the hierarchical model.

- Ground truth (field-measured DBH, applied to the same Verra chain).
- Hierarchical model posterior mean (Jensen-corrected): +4.5%.
- Hierarchical model posterior p_{25} : +1.0%.
- Hierarchical model posterior p_{10} : −2.3%.
- F4F’s 80% rule applied to the mean: −16.4%.

6.3 Why a calibrated quantile is better

The 80% rule is a fixed multiplicative haircut: it does not adapt to plot composition, model uncertainty, or species heterogeneity. A posterior lower-quantile estimate has three properties the haircut does not:

- It adapts: plots with low predictive uncertainty (uniform species, narrow age distribution, abundant training data for those species) get a tighter haircut; plots with high uncertainty get a larger one. The 80% rule cannot distinguish these.
- It is interpretable: “with 75% posterior probability the true plot CO_2 is at least the reported value” is a statement an auditor can verify.
- It is tighter at equivalent audit-safety. Our p_{25} estimate is closer to ground truth in absolute terms (+1.0% vs. −16.4%) while keeping the conservative side of the calibration imbalance, since the hierarchical model is already slightly under-confident in the audit-favourable direction.

This recovery is real money. At a representative voluntary-market carbon price of \$30/tCO₂, the gap between the 80% rule and a calibrated p_{25} on the test set is approximately 24.8 tCO₂, or about \$750. Scaled to a 1,000-hectare project, the same gap is on the order of \$100,000–\$500,000 per cycle, depending on plot composition. (These are illustrative back-of-envelope figures; project-level deployment would warrant per-plot quantile calibration.)

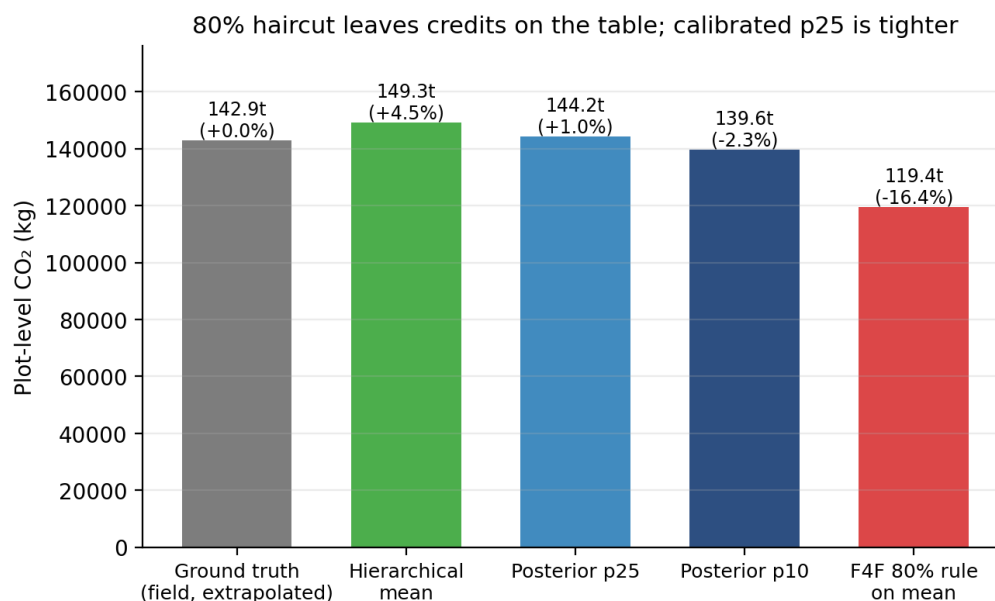


Figure 6: Plot-level CO₂ on the test set under five reporting policies. The 80% haircut on the mean (red) under-reports by 16.4%; a calibrated posterior p_{25} (light blue) under-reports by 1.0% with the same audit-favourable direction.

7 Discussion

7.1 What partial pooling resolves from TreeLens’s *aleatoric uncertainty* framing

TreeLens (Sections 5.1, 5.3, Figure 3, Table 3) attributes much of the within-plot DBH variance to *aleatoric uncertainty* — unobservable, irreducible variation.

Our reading is more optimistic: a substantial fraction of this *aleatoric* variation is in fact *epistemic*, attributable to species and age, which are observable and present in the data. The hierarchical model reduces test RMSE by approximately half by exploiting exactly those covariates. What TreeLens describes as irreducible noise is, in part, the cost of choosing a species-agnostic modeling target. The remainder is genuine measurement noise (saplings in particular have a measurement floor of ± 10 –20% relative to true DBH; see Section 7.2) and within-species genetic / micro-site variation that no model can resolve from (TC, TH) alone.

7.2 Limitations

Saplings. Trees below 5 cm DBH carry $\sim 1\%$ of plot CO₂ in our test set but show +25% bias on the DBH scale. This is not a modelling problem: tape measurement of a ~ 2 cm trunk under field conditions has 10–20% relative noise. The bias is downstream-irrelevant for VMD0001 plot totals, but should be flagged in any per-tree report.

Large trees. The test set contains only two trees above 50 cm DBH. Statements about model behaviour at very large DBH are not supported by this evaluation. Production deployment in mature forest plots would warrant separate validation.

Orchard age slopes. Five intensively-managed orchard species (Guava, Kaju, Lemon, Orange, Chiku) showed negative posterior age slopes in an earlier model iteration, biologically implausible if interpreted at face value. The most likely explanation is data leakage from an unobserved confounder: older orchards receive more intensive pruning, which holds DBH growth

back. We address this with a non-negativity constraint on the species-level age slope; this preserves biology but means the constrained species are now informed mostly by the prior on age rather than the data. A more principled fix would explicitly model management intensity, which the public dataset does not currently include.

Geographic scope. The dataset is exclusively Maharashtra semi-arid agroforestry. The hierarchical model has no information about other climates, soils, or species mixes; performance outside this scope is unknown.

7.3 Future work

The natural next steps are: (i) replacing the 80% rule with a project-level calibrated quantile, validated on a held-out plot set; (ii) species-specific allometry (where available) replacing the single Verra equation, which the TreeLens paper also flags as a limitation; (iii) explicit modelling of management intensity for orchard species to disentangle the age/pruning confound; (iv) errors-in-variables modelling for the sapling bin, if per-tree reporting at small DBH becomes operationally important; and (v) cross-geography validation as the dataset expands beyond Maharashtra.

Code and data

The hierarchical model implementation, evaluation pipeline, and figure-generation scripts are open. The ground-truth measurements used for training and evaluation are from the public F4F datasets repository [5] (13 October 2025 release), used without modification.

References

- [1] A. Gupta, M. Datar, P. Mulay. *AI Based Automated Calculation of Carbon Stock in Agroforestry Plantations* (TreeLens White Paper v1.0). Farmers for Forests, August 2025. <https://www.farmersforforests.org/FFF/images/our-tech/tree-lens-white-paper.pdf>
- [2] Verified Carbon Standard. *VMD0001: Estimation of carbon stocks in the above- and below-ground biomass in live tree and non-tree pools (CP-AB), v1.2*. Verra, 2022. <https://verra.org>
- [3] O. Abril-Pla, V. Andreani, C. Carroll, et al. *PyMC: A modern and comprehensive probabilistic programming framework in Python*. PeerJ Computer Science 9:e1516, 2023.
- [4] G. W. Sileshi. *A critical review of forest biomass estimation models, common mistakes and corrective measures*. Forest Ecology and Management 329, 237–254, 2014.
- [5] Farmers for Forests. *Public datasets repository*. <https://github.com/Farmers-For-Forests/public-datasets>. Includes the 13 October 2025 ground-truth release used in this work.

A Appendix: per-species posterior parameters

Table 2: Posterior medians of species-level parameters. *age slope** marks species whose age feature is masked because within-species age CV < 0.10 — their printed age slope is the prior mean, not a learned quantity.

Species	Baseline DBH (cm)	Crown slope	Crown curve	Height slope	Age slope	Noise σ
Amla	8.41	0.207	0.066	0.421	0.400	0.157
Bakul	6.85	0.299	-0.007	-0.224	0.734	0.145
Chandan	6.03	0.481	-0.026	0.317	0.171*	0.144
Chiku	8.82	0.345	0.187	0.259	0.033	0.258
Custard apple	8.11	0.515	-0.048	0.342	0.004	0.197
Guava	7.01	0.391	0.133	0.212	0.016	0.259
Jamun	9.71	0.344	0.036	0.507	0.261	0.213
Kaduneem	7.55	0.445	0.123	0.456	0.197	0.185
Kaju	6.88	0.370	0.055	0.324	0.176*	0.298
Kanchan	8.54	0.396	0.017	0.177	0.112	0.179
Kashid	8.06	0.224	-0.009	0.178	0.049	0.240
Lemon	10.27	0.580	-0.080	0.377	0.009	0.258
Mango	9.64	0.559	-0.005	0.209	0.335	0.236
Orange	9.94	0.493	-0.105	0.449	0.010	0.234
Pimpal	8.01	0.278	0.018	0.455	0.052	0.194
Sandalwood	7.99	0.385	-0.122	0.204	0.114	0.164
Shisam	8.63	0.294	-0.028	0.222	0.269	0.192
Tamarind	8.35	0.525	0.040	0.447	0.102	0.256